

NE9 Coursework 2: Fast Reactors

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1 Time-dependence of feedback

- a) The plot below shows a trace of power and reactivity following the PFR rod drop on 29 March 1985. The power exhibits a step change with a certain magnitude. Calculate the power change we would expect and compare it to what was observed. Ignore the subsequent evolution of power with time.

From observing the figure, it can be seen that the power jumps from around 546 MW to around 488 MW. Therefore the observed change in power ΔP_{obs} is given by:

$$\begin{aligned}\Delta P &= 546 \text{ MW} - 488 \text{ MW} \\ &= \boxed{58 \text{ MW}}.\end{aligned}\tag{1}$$

It can also be observed that the rod drop causes a reactivity drop $\Delta \rho_{\text{rod}}$ of around:

$$\Delta \rho_{\text{rod}} = -11.1 \text{ } \epsilon.\tag{2}$$

Therefore, the expected change in power ΔP_{exp} is:

$$\begin{aligned}\Delta P &= \left(1 - \frac{1}{1 - -0.111}\right) 546 \text{ MW} \\ &= \boxed{54.6 \text{ MW}}.\end{aligned}\tag{3}$$

This result underestimates the power drop by 3.4 MW or 5.86% compared to 58 MW.

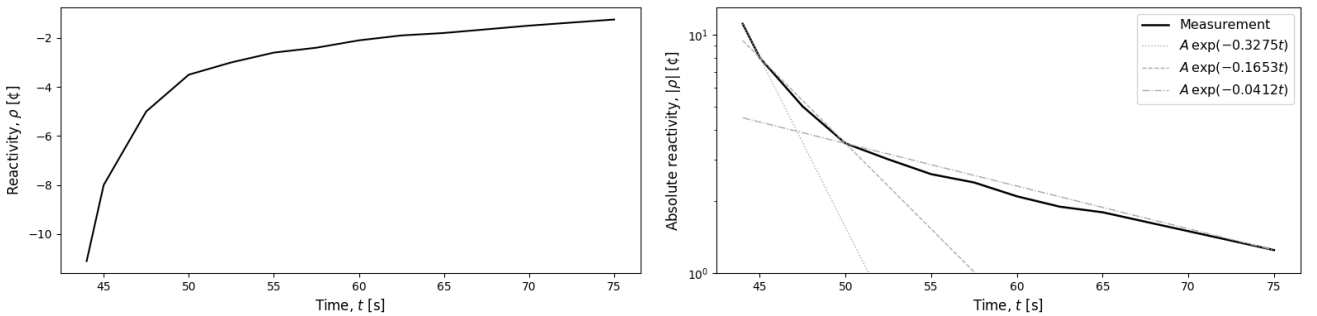
- b) After the change in power, the reactivity begins to recover due to feedback. The shape is not quite exponential, suggesting several contributing effects. Plot the logarithm of reactivity over time and estimate the time constants for the contributing effects. Based on these time constants, explain which feedback phenomena they might be associated with.

The first step for the creation of the log-linear plot, was to create extract the numerical data from the graph given in the assignment. This resulted in Figure 1a (for exact numbers refer to Table 2). Changing the y -axis to a logarithm resulted in Figure 1b.

Three roughly linear regions could be observed in Figure 1b: 44 s to 45 s, to 50 s, and 50 to 75 s. These linear regions correspond to exponential relations, which are also shown in the figure. The time constants are:

$$\begin{aligned}k_1 &= 0.3275 \text{ s}^{-1} \\ k_2 &= 0.1653 \text{ s}^{-1} \\ k_3 &= 0.0412 \text{ s}^{-1}\end{aligned},\tag{4}$$

where k_1, k_2, k_3 denote the respective time constants for the 3 regions.



(a) Linear plot of the reactivity.

(b) Log-linear plot with exponential relationships.

Figure 1: Plots of the reactivity over time after a rod drop, with data obtained using the graph in exercise 1. The gray lines in the log-linear plot depict the exponential relationships found for 3 different sections of the plot, as shown in the legend.

The magnitude of each of the coefficients is inversely proportional to their time scale. Thus, the coefficients k_1, k_2, k_3 represent increasingly slow feedback effects. The specific feedback mechanism can be determined by their time scale:

- $1/k_1 \approx 3$ s. This corresponds to the fastest-acting feedback: **Doppler broadening** (or in this case narrowing) of the absorption peaks. When the fuel temperature decreases, the thermal motion the fuel atoms decreases in the reference frame of the neutrons. This effectively narrows the absorption peaks and increases reactivity.
 - $1/k_2 \approx 6$ s. The next fastest feedback comes from the **bowing effect** of the sub assemblies. The PFR sub-assemblies have a ‘cantilever’ mounting configuration: they only have a single mounting point¹. When the temperature is high, the sub-assemblies have a tendency to bow outward. This increases the distance between the fuel and worsens the neutron economy. It follows that a decrease in temperature means that the fuel assemblies will not bow outward as much, increasing the reactivity.
 - $1/k_3 \approx 24$ s. This is the slowest feedback and therefore must correspond to the mechanism with the largest thermal mass. This feedback comes from the **thermal contraction of the entire core**. When the core temperature drops, the thermal contraction means that the core is physically smaller and the coolant is more dense. This softens the spectrum and therefore decreases neutron leakage. The improved neutron economy increases reactivity.
- c) *Following this step change, the reactor conditions will vary through time. The previous part of the question was concerned with the short-term response of the reactor. Now use the quasi-static approach to determine the long-term change in the reactor operating conditions. (...) Plot the evolution of the power and the reactor inlet and outlet temperatures with time. Do not worry about actual values of time, just show the behaviour. For clarity you may assume some typical coolant temperatures for the start of the transient. Again, ignore the power and reactivity trace on the PFR results for this analysis.*

At equilibrium, the total reactivity must be 0. Therefore, the reactivity induced by the rod must be fully canceled by the feedback mechanisms:

$$-\Delta\rho_{\text{rod}} = A \frac{\Delta P_{\infty}}{P_0} + B \Delta[P/F]_{\infty} + C \Delta T_{\text{in}}, \quad (5)$$

where the subscript 0 denotes the state before the drop of the rod and ∞ is the final equilibrium state. There is no way to know ΔT_{in} without information about the secondary side. This is not a problem, since C is very low, and ΔT_{in} is typically on the order of 1 K. Thus, it is valid to assume a constant temperature, $\Delta T_{\text{in}} = 0$. Furthermore, the flowrate F stays constant. Rewriting the equation gives:

$$-\Delta\rho_{\text{rod}} = \Delta P_{\infty} \frac{A}{P_0} + \Delta P_{\infty} \frac{B}{P_0}. \quad (6)$$

Solve for ΔP_{∞} .

$$\begin{aligned} \Delta P_{\infty} &= -\frac{P_0 \Delta\rho_{\text{rod}}}{A + B} \\ &= -\frac{546 \text{ MW} \cdot -0.111 \$}{-0.5 \$ + -0.4 \$} \\ &= -67 \text{ MW}. \end{aligned} \quad (7)$$

This results in a final power of $P_{\infty} = 479$ MW. As for T_{out} , considering that the flow stays constant, the change in $T_{\text{out}} - T_{\text{in}}$ must be proportional to P . Therefore:

$$T_{\text{out},\infty} = T_{\text{in},\infty} + \frac{P_{\infty}}{P_0} (T_{\text{out},0} - T_{\text{in},0}). \quad (8)$$

Assume that $T_{\text{out},0}$ and $T_{\text{in},0}$ are equal to their reference temperatures², so $T_{\text{in},0} = 687$ K and $T_{\text{out},0} = 835$ K. Further, assume an inlet temperature drop of 1 K, so $T_{\text{in},\infty} = 686$ K. The final outlet temperature becomes:

$$T_{\text{out},\infty} = 815 \text{ K}. \quad (9)$$

The plot is shown in Figure 2. To summarize, this plot consists of:

- For P : a step response from 546 MW to 488 MW, followed by an exponential decay towards 479 MW.
- For T_{in} an exponential decay from 687 K to 686 K.
- For T_{out} an exponential decay from 835 K to 815 K.

¹A.G. Frame, W.G. Hutchinson, J.M. Laithwaite, H.F. Parker, *Design of the Prototype Fast Reactor*, Editor(s): P.V. EVANS, Fast Breeder Reactors, Pergamon, 1967, Pages 291-315, ISBN 9781483198415, <https://doi.org/10.1016/B978-1-4831-9841-5.50019-4>

²S. E. Jensen, P. L. Oelgaard and Nordisk Kernesikkerhedsforskning, Roskilde (Denmark), *Description of the prototype fast reactor at Dounreay*, Dec. 1995, <https://inis.iaea.org/records/jnwwf-f7j05>

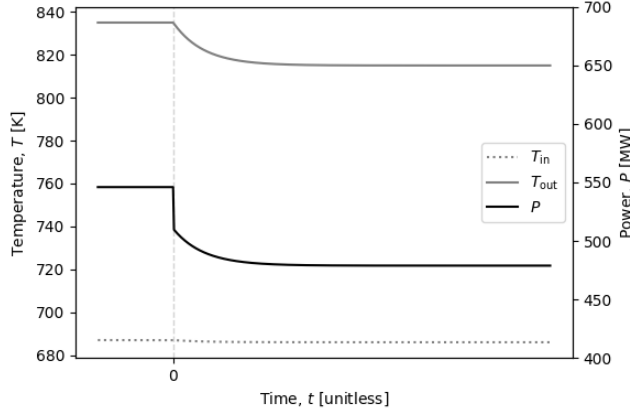


Figure 2: Plot of the quasi-static response of the PFR rod drop.

d) Now imagine that the inlet coefficient somehow has the opposite sign. What would be the long-term behaviour of the core under the same transient conditions?

Before considering C , it is useful to analyze the other feedback mechanisms. The reactivity contributed by A and B are given by:

$$\begin{aligned} (A + B) \frac{\Delta P}{P_0} &= (-0.5 \$ - 0.4 \$) \cdot \frac{58}{546} \\ &= -0.096 \$, \end{aligned} \quad (10)$$

where ΔP is the power change immediately after the drop. The behavior of the reactor is entirely dependent on if the contribution of C is enough to negate this negative feedback. The critical value of ΔT_{in} when this happens can be round by equating:

$$\begin{aligned} C \Delta T_{in} &= -0.096 \$ \\ -0.005 \$/K \cdot \Delta T_{in} &= -0.096 \$ \end{aligned} \quad (11)$$

Solving for ΔT_{in} gives:

$$\Delta T_{in} = 19.2 \text{ K}. \quad (12)$$

Thus, there are two different scenarios depending on the magnitude of the temperature drop:

1. $\Delta T_{in} > 19.2 \text{ K}$: Admittedly, such a large temperature drop is extremely unlikely. Regardless, assume that this is the case. The overall reactivity feedback would become positive, so the reactor would be unstable. Therefore, the negative reactivity caused by the rod drop would not be counteracted, but rather exacerbated by the temperature feedback. The reactivity would become more and more negative until the reactor power drops to 0.
2. $\Delta T_{in} < 19.2 \text{ K}$: This scenario is much more likely. The core would remain stable and eventually settle at some equilibrium power. The fact that C is positive would mean that the power drop would be larger than the one determined in (7). However, this does not mean that a positive C is desirable in any way. The inlet temperature may rise due to reasons other than a change in P , such as a fire. A positive reactivity feedback would be dangerous in that case.

2 Critical Enrichment: Using the CSGROV code set up a diffusion-theory model of a cylindrical fast reactor core.

a) Determine the enrichment x to make the core critical ($k_{eff} = 1$).

Assuming a homogeneous core, the critical level of enrichment x was determined at:

$$x = \boxed{12.48\%}, \quad (13)$$

which results in a multiplication factor of $k = 1.0000$.

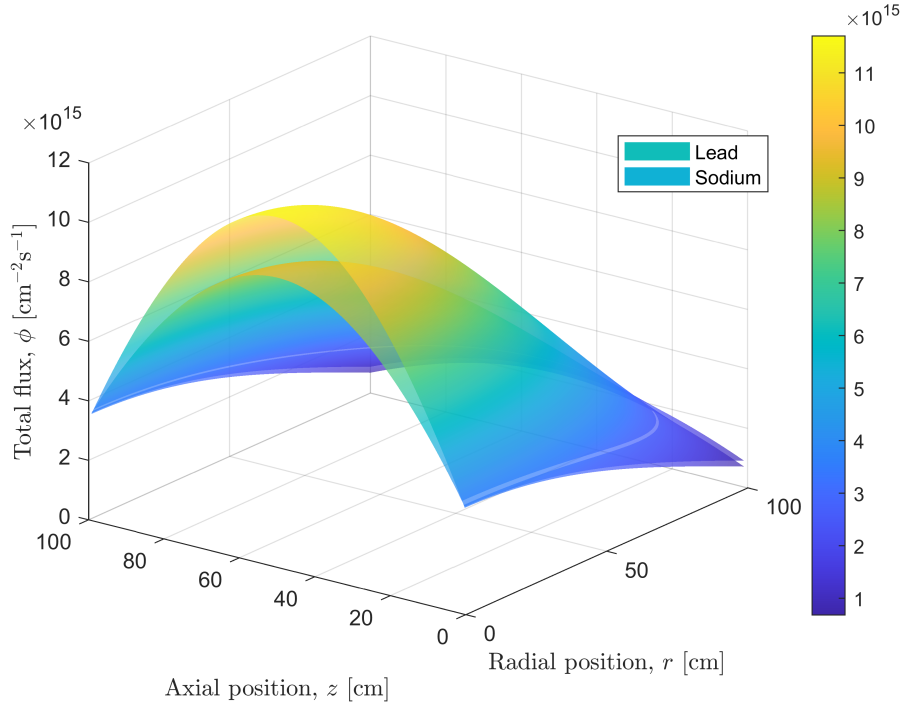


Figure 3: Surface plot of the neutron flux for a critical fast reactor $k \approx 1$, simulated with CSGROV. The upper surface is cooled by lead, the lower surface is cooled by sodium.

- b) Replace the sodium coolant with lead (density 11.3 gcm-3, atomic weight 207). Determine the critical enrichment. Explain the difference between lead and sodium, with reference to results for the flux provided by CSGRV.

Changing the coolant to lead, the critical level of enrichment x changes to:

$$x = \boxed{11.88\%}, \quad (14)$$

which results in a multiplication factor of $k = 1.0002$.

The cause of this lower level of enrichment becomes clear when looking at the neutron flux depicted in Figure 3. The lead reactor clearly has more neutron flux. The reason for this is two-fold: First, lead is much heavier than sodium, so its capture cross-section is lower; second, lead is a worse moderator, which leads to a harder spectrum and a higher neutron yield. The combination of these two effects means that lead-cooled fast reactors need less enrichment for to stay critical.

3 Sodium Void Reactivity: Divide the core model of exercise 3 (sic.) into two zones with a central region of $r < 0.065\text{m}$. The inner zone is an approximation for the central fuel subassembly.

- a) Estimate the reactivity worth of the sodium in the central subassembly by removing the sodium from the inner zone and, without changing the enrichment, calculating k_{eff} .

The CSGROV simulation of the core with the central subassembly removed, gives:

$$k_{\text{eff}} = 1.0004, \quad (15)$$

which corresponds to a reactivity of:

$$\begin{aligned} \rho &= \frac{k_{\text{eff}} - 1}{k_{\text{eff}}} \\ &= \frac{1.0004 - 1}{1.0004} \\ &= \boxed{0.0004}. \end{aligned} \quad (16)$$

- b) Estimate the reactivity worth of all the sodium in the core by removing it and calculating k_{eff}

Removing all the sodium from the core gives:

$$k_{\text{eff}} = 1.0124, \quad (17)$$

which corresponds to a reactivity of:

$$\begin{aligned}\rho &= \frac{k_{\text{eff}} - 1}{k_{\text{eff}}} \\ &= \frac{1.0124 - 1}{1.0124} \\ &= \boxed{0.0123}.\end{aligned}\tag{18}$$

- c) *Explain the results of (a) and (b) with reference to the effect of sodium on reactivity at different locations in the core*

This question is referring to a peculiarity between the results in (16) and (18). The result in (18) was obtained by removing $85^2/6.5^2 = 171$ times as much sodium from the reactor compared to the result in (16). If the effect of sodium was homogeneous throughout the core, one would expect a reactivity change of $171 \cdot 0.0004 = 0.0684$. However, the result in (18) is more than 5 times lower. Clearly, the sodium in the center of the core has much more impact on reactivity than the sodium in the periphery.

The reason for this discrepancy comes from the way that sodium effects the reactivity. As explained in subsection b), sodium worsens the neutron economy by capturing and moderating neutrons. These interactions are the reason why sodium reduces reactivity. Importantly, both these effects are proportional to the local neutron flux. If the neutron flux is high, the reduction in neutrons by adding sodium is also high. Therefore, the reactivity worth of sodium is proportional to the local flux. Looking at Figure 3, it is clear that the flux is highest in the center of the core. This is why removing sodium from the center has a greater impact on the reactivity.

- d) *We would like to make the full-core sodium void coefficient less positive. Experiment with the overall size and aspect ratio of the core to show how this could be achieved.*

The relationship between the core geometry and the sodium worth was studied by varying the radius and height of the core, and comparing the resulting difference in reactivity. The results are shown in Table 1.

From the first 6 rows, it is clear that larger cores increase the full-core sodium void coefficient. Regarding the ratio, the bottom 3 rows have the same volume. It can be seen that the full-core sodium void coefficient is higher for longer axial heights. In conclusion, the best way to reduce the full-core sodium void coefficient is by reducing the size of the core, and by reducing the axial height compared to the radius.

Table 1: CSGROV results for different core geometries with and without sodium coolant. The first two columns are the core geometry. The third column is the k_{eff} result with a sodium coolant. The fourth column is the k_{eff} with all the sodium removed. The fifth column is the resultant difference in reactivity. The highlighted row is the default core geometry from exercise 2. The bottom 3 rows have equal volume.

Radius [cm]	Height [cm]	Sodium k_{eff}	Void k_{eff}	$\Delta\rho$
80	100	0.9727	0.9808	0.0085
80	110	0.9882	0.9985	0.0104
85	100	0.9842	0.9942	0.0102
85	110	1.0000	1.0124	0.0123
85	120	1.0131	1.0276	0.0139
90	110	1.0104	1.0247	0.0138
69.71	200	1.0208	1.0376	0.0159
90	120	1.0238	1.0403	0.0155
103.92	90	0.9965	1.0098	0.0132

- 4 **Temperature Coefficient of Reactivity:** *Calculate the effect of increasing the temperature of the core and coolant by 100 K for the reactor dimensions in Exercise 2. Increase the linear dimensions (core radius and height) without changing the mass of fuel and structure in the core (i.e. by adjusting the densities to compensate for the increased volume). Change the density of the sodium appropriately.*

It is apparent that the volumetric expansion coefficient of sodium is higher than that of stainless steel. For this question, I will be assuming that the core is strong enough, such that the increased pressure from the sodium has a negligible effect on the volume of the core. This is a reasonable (and standard) assumption, considering that the core is made of steel.

a) Estimate k_{eff} for the expanded core.

Using the same enrichment level as found in (13) results in a multiplication factor of:

$$k_{\text{eff}} = \boxed{0.9988}. \quad (19)$$

b) Deduce an approximate value for the temperature coefficient of reactivity in c/K .

The change in reactivity is given by:

$$\begin{aligned} \Delta\rho &= \frac{k_{\text{eff}} - 1}{k_{\text{eff}}} \frac{100}{\beta} \text{ c} \\ &= \frac{0.9988 - 1}{0.9988} \frac{100}{0.003} \text{ c} \\ &= -40.05 \text{ c}. \end{aligned} \quad (20)$$

Therefore the temperature coefficient of reactivity is:

$$\begin{aligned} \alpha_T &= \frac{\Delta\rho}{\Delta T} \\ &= \frac{-40.05 \text{ c}}{100 \text{ K}} \\ &= \boxed{-0.4005 \text{ c/K}}. \end{aligned} \quad (21)$$

5 Role Play: Choose a country with a nuclear power programme (e.g. UK, USA, France, China, India, Russia, ...). Imagine you are the scientific advisor to the government minister for energy. He or she has to decide whether to invest \$500M in the development of fast reactors and has asked your advice. Write an outline briefing paper explaining the advantages and disadvantages. You might remain impartial overall, but if you decide either for or against state your reasons.

For this assignment I will be using € instead of \$, because that is the official currency of the Netherlands. If you are interested: \$500 $\approx$ €433. I don't think the exact figure is very important so I will be rounding that up to €500. The letter is on the next page.

To the Energy Minister of the Netherlands,

Regarding your proposed investment of €500M for the development of a fast-neutron reactor (FNR), I must say that I am delighted by the prospect. FNRs have several advantages (including increased efficiency and others I will discuss shortly) which warrant further study. This is not to mention the positive externalities from the technological spillover of any major scientific undertaking. However, before I give my wholehearted support to this investment, I feel obligated to clear up some common misconceptions you may hold.

The first misconception is that FNRs are safer than traditional reactors. In actuality, their safety benefit is uncertain and limited to specific accident scenarios. More importantly, the statement seems to imply that regular reactors are unsafe. They are not. In its 70+ year existence, nuclear power has consistently maintained its position as one of the safest forms of electricity. Its deaths per kWh statistic rivals that of solar. If this investment is intended to improve public safety, I would urge you to build more green energy sources instead. The pollution from our coal plants kills orders of magnitude more people than nuclear ever has.

Somewhat related to this topic is the idea that fast reactors produce less nuclear waste. Indeed, FNR waste does have a much shorter lifespan than traditional reactors. However, this assumes that waste disposal is currently a problem without a solution. But a solution does exist, and it is miles cheaper than developing a new reactor: deep geological repositories. If your concern is waste, perhaps it would be a better idea to follow our southern neighbors and actually establish a task force to research long-term waste disposal. Instead, you are delaying it to “some time before 2130”.

Another oft-cited benefit of fast reactors is their ability to breed fuel. This is another useless advantage. Fuel only makes up about 15% of the levelized cost for a reactor. Clearly, the economic benefit of recycling is minimal, especially considering the cost of developing an FNR. Perhaps, instead, the goal is energy sovereignty. However, the EU already has plenty of known uranium deposits. This is not to mention that achieving true autarky is delusional for a country as small as ours.

Truthfully, €500M is a laughable amount of money to develop an entirely new reactor type. The cost of constructing nuclear power stations is in the order of tens of billions. In that sense, this is clearly more of a research grant, which is admittedly something I would never oppose; science is a productive force after all. However, it was your cabinet that recently cut the budgets of our universities by over €25B. I happen to know that part of this money went to the excellent Delft University Reactor, whose researchers were actively working on fast reactors. It makes me wonder if this is an earnest investment in green energy technology or merely a cynical attempt to distract the public from your broken election promise: to build new green energy.

Sincerely,
Andy Zhang

500 words

APPENDIX: exact data

Table 2: Data used for the creation of the graphs in exercise 1a.

t	ρ [c]
44	-11.1
45	-8
47.5	-5
50	-3.5
52.5	-3
55	-2.6
57.5	-2.4
60	-2.1
62.5	-1.9
65	-1.8
70	-1.5
75	-1.25